

VideoGlaze: A Tool for Combating Style-Mimicry in AI Generated Videos

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Abstract—AI generated videos have become increasingly popular in recent years, including videos of fake actors, and AI-generated animation. There are a number of tools that have been proposed to cloak real photos and videos from deepfakes, but few to prevent stylistic mimicry for art, and only for images. Our goal is to apply a pre-existing tool for anti-AI style mimicry in images to videos. Users of this novel tool could apply it to their own animations and art videos to protect them from AI generation tools. To explore this topic further, we are looking to apply the anti-AI image generation tool Glaze to combat AI art video generation. Specifically, we would like to explore if different video generation models respond differently to defensive perturbations. We would like to evaluate whether Glaze-based protections are consistently effective across multiple systems or maybe whether some models are able to bypass or ignore these perturbations. Our goal is to assess the robustness and transferability of defensive techniques in the video generation area. Although Glaze cannot be used retroactively on images that have already been uploaded to the internet, it can be used proactively moving forward for new images that people will post online, and we are seeking to extend this future protection to videos. Alongside further researching privacy preservation tools for AI imaging and text generation, we hope to empirically test our topic by experimenting with anti-AI image tools on the market.

Index Terms—Artificial Intelligence (AI), Generative AI, AI Art, Privacy Protection

I. INTRODUCTION

Generative artificial intelligence is rapidly transforming digital content creation, specifically in image and video synthesis. The most recent advancements in image diffusion based models have allowed users to generate hyper realistic images, animations, GIFS, videos, and more from a single text prompt. These AI tools can create stylized videos that approach similar quality to human artists which proponents argue can lower barriers to artistic multimedia production. However, they also raise major concerns around intellectual property, artistic consent, and data privacy. In this paper we focus on one such concern: style mimicry, where generative AI models are able to replicate the visual style of an artist without their permission.

This issue first garnered attention within the image generation domain, where text-to-image models such

as Stable Diffusion and Midjourney demonstrated the capability to imitate well known artists' styles. In 2023, three visual artists sued Stability AI, Midjourney, and DeviantArt for alleged copyright infringement after their work was included in the LAION dataset which consists of 5 billion images scraped from the internet used by these companies to develop their AI image generators [1]. The plaintiffs argued that the AI image generators not only used their work without permission nor compensation, but also that the tools could be used to replicate their own art styles to generate similar work.

Thus, the consequences of style mimicry extend beyond claims of copyright violations. Many artists argue that generative AI tools threaten their economic stability as well since automated replications of their styles, ones which took years to develop, are being copied and potentially sold within minutes [1]. In many cases, only a small number of publicly available works are required for such reproduction, creating a very low bar for models to copy artistic style. This problem especially has resulted in demand for technical and ethical defenses against unauthorized style replication.

Many tools have been proposed and developed to combat copyright-infringement in AI generative models, including machine unlearning [2, 3], watermarking [4], and anti-copyright prompting [5]. However, many of these tools are proactive, requiring AI model developers to implement them during the model development or generation process. Others are limited in efficacy, offering no guarantee of protection for creatives [5]. Few defensive tools, implemented by users without needing to interact directly with model parameters, exist for preventing copyright infringement in video generation. Most focus on detection [4] or on preventing photorealistic deepfakes rather than stylized videos and animation [6]. Meanwhile, generative AI capabilities are advancing towards video production. Some modern text-to-video systems can generate animated clips and cinematic scenes that closely resemble professional work in the fields of animation, character design, and motion capture [7].

Our project explores whether image based anti-style mimicry tools can be adapted to protect video content

from generative AI systems as well. One tool, Glaze, is applied to images defensively to prevent AI models from learning and mimicking the artist’s unique style [8]. While Glaze has proven effective for static images, we propose a new tool, VideoGlaze, that expands its usage to video media and animated content and can be found. Specifically, we investigated whether applying Glaze to individual frames of animation clips can reduce AI models’ ability to successfully imitate an artist’s style in text-to-video models. We focus on style mimicry because these models are typically trained on large open source datasets, which means artists may not know that their creative work is being stolen. The benefit of this is that Glaze can be used defensively, protecting artists from AI without them ever having to engage directly with the model.

A. Research Questions

1) *RQ1*: What current protections exist for preventing video style mimicry, and what gap does our frame-by-frame Glaze approach address?

We investigate existing defenses related to image and video stylistic copying prevention II.

2) *RQ2*: How effective is VideoGlaze at preventing style copying by text-to-video generators?

We evaluate whether our proposed method effectively reduces style imitation according to three empirical evaluation metrics: CLIP similarity, perpetual similarity, and artistic style representations V VI.

3) *RQ3*: What limitations exist in applying image based defenses to video content and what future research directions are needed?

We analyze weaknesses and limitations to our approach, as well as future directions for improving VideoGlaze and our evaluation metrics VII.

II. BACKGROUND

Our paper extends Glaze, a novel defensive tool designed to protect art from being exploited by AI generative models. Glaze applies nearly invisible adversarial perturbations to artwork before it is published online. These alter how machine learning systems interpret the art style of the cloaked image (after Glaze is applied) while the actual artwork remains visually unchanged to human viewers. Figure 1 shows how Glaze’s process works on static images. Glaze computes style-specific features of an artwork then adds noise perturbations to those specific features. The art style is identified by leveraging “style transfer,” a computer vision technique which transforms an initial image into one with a different style. In an initial review, Glaze achieved over a

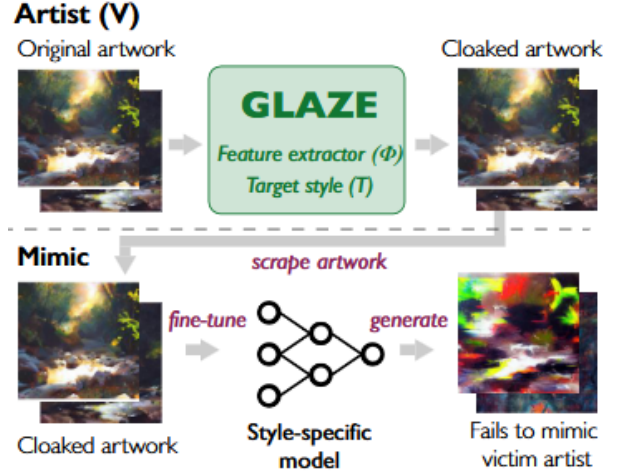


Fig. 1: Overview of Glaze’s process: **(Top)** An artist V applies the Glaze cloaking algorithm to generate cloaked versions of V’s art pieces. **(Bottom)** A mimic scrapes the cloaked art pieces from online and uses them to fine-tune a model to mimic V’s style. When prompted to generate artwork in the style of V, mimic’s model will generate artwork in the target style T, rather than V’s style [8].

92% success rate in disrupting mimicry attempts under standard settings [8]. Shan et al. have also developed Nightshade, a tool similar to Glaze that poisons models that attempt to use cloaked images in their training data [9].

III. VIDEOGLAZE OVERVIEW

VideoGlaze is an open-source software that takes an input video and returns a cloaked video that appears identical to a human viewer but confuses an AI model that tries to learn its stylistic features.

First, each animation clip is decomposed into its individual image frames. Each frame is then individually processed through the regular Glaze algorithm for image cloaking. The cloaked frames are reconstructed into their original order to form a fully cloaked animation sequence. Since each frame is cloaked independently, it is possible that slight perturbation differences may introduce problems like flickering or inconsistencies when the frames are combined into a video format. Additionally, applying the full Glaze optimization to each video frame would take hours per minute of footage. To address this challenges, we implement two additional features to optimize VideoGlaze’s process:

- **Temporal gating:** Frames that are visually near-identical to the previous frame skip the optimizer entirely and re-use the previous perturbation. This

eliminates redundant work in static shots and slow pans.

- **Warm start:** Instead of initializing the perturbations from zeros for each frame, the previous frame’s perturbation is used as the starting point. Frames in a video are typically very similar, so the optimizer starts close to a good solution and converges in far fewer iterations.

These features help reduce the total time to cloak a full animation by using the previous frames’ perturbation results as a starting point (warm start) and skipping the iterations entirely if frames are close enough to their predecessor (temporal gating). By repeating previous frames’ optimization and using warm start for similar frames, we also reduce the distortion that VideoGlaze may cause to adjacent frames that would disrupt animation flow.

VideoGlaze is intended for individual artists to protect their animated clips, gifs, and other artistic videos from style mimicry by generative AI models. After cloaking with VideoGlaze, every frame of the video is protected, so that if a bad faith actor inputs individual frames from a cloaked video into a text-to-video generator as references images, the resulting output cannot successfully mimic the style of the video. VideoGlaze also protects against less explicit use cases by providing this same protection when cloaked videos are used in AI model training data without an artists’ knowledge or consent. Given how important online audiences and social media are to modern artists, this tool helps protect their future work uploaded online from being misused.

IV. METHODOLOGY

To evaluate whether image based anti-style mimicry defenses can be extended to video content, we designed an experimental framework that applies Glaze frame by frame to short animation clips and measures whether this protection helps reduce style mimicry and replication in three online text-to-video AI generators. Since our project focuses more on defensive tooling for animators, our experiment setup aims to simulate how an artist might protect their original work before publishing it online.

A. Video Selection

We compiled a dataset of short animation clips sourced from recent open-source repositories and publicly available datasets. Our focus was on recently uploaded animation clips that likely would not have been included in prior training data for any of the video generation services we used to test VideoGlaze. We selected six GIF-length animation clips, each in the range of 21 to 75 frames long. We intentionally

selected shorter clips because they are computationally easier to process while still preserving the temporal information necessary to evaluate style consistency for videos. The short clips allow us to test multiple text-to-video generation models faster.

Additionally, the selection animations were selected for diversity across key attributes for style mimicry. First, we varied the release year (date posted online) from 2020 to 2026. Since older clips are more likely to have appeared in previous training data sets, this adds an additional metric to test if VideoGlaze appears less effective on clips with a high likelihood of inclusion in training data. In theory, VideoGlaze should work even if videos were included in training datasets since it cloaks the video as to make the animation style unrecognizable. Thus we include newer and older clips to verify this condition. We also selected across different animation styles (hand drawn, 2D, cartoonish vs realistic) to test whether these factors influence the model’s ability to replicate style.

B. VideoGlazing Process

For our trials, we ran VideoGlaze with the following Glaze parameters: `perturbation budget=0.1` and `iterations=500`. The perturbation budget defines the additional noise threshold which Glaze uses to cloak the image so that AI models confuse its style. This was a relatively high perturbation budget (0.05 is imperceptible to more people, which 0.1 to 0.2 offers stronger protection) [8] which we selected because we wanted our images to be as protected as possible without being largely noticeable to human viewers. We could not go much higher than 0.1 since the GIFS were relatively small, and higher perturbations would have been more noticeable. The number of iterations defines how many optimizer steps Glaze runs; fewer iterations converge faster but result in weaker cloaking. We selected 500 to match the original Glaze paper [8]. Defaults for used for the other parameters.

C. Text to Video Model Testing

After applying VideoGlaze, we submitted each cloaked video to the AI text-to-video models Pika Labs [10]. This models was chosen primarily for ease of access, and also because it specifically focuses on generating videos in specific artistic styles, allowing for reference style images as well. This tools takes in an initial and final reference frame as well as a text prompt. We submitted the original and cloaked initial and final frames to each AI model, with the prompts outlined in Table ???. Although our tool protects the entire video, we assume that a bad faith actor could take any frames from an artists’ online animation and submit them to a similar

Test Video	# Frames	AI Prompt
Lava lamp	21	Animate the lava in this lava lamp.
Eye	24	Animate this eye to open and close.
Dynamite	21	Animate this dynamite to explode into a ball of fire.
Leprechaun	26	Animate this leprechaun to do an Irish dance.
Waves	48	Animate these waves to move.
Turtles	75	Animate these turtles to swim forward.

TABLE I: List of test videos, their frame length, and the prompts used to generate mimicry videos with Pika. For all the videos tested, see here.

text-to-image model. Thus we assume that the first and last frames are representative of any two frames from a source video, and will work similarly against these misuses that VideoGlaze aims to protect against.

V. EVALUATION METRICS

We evaluate each generated clip against its reference clip with three complementary frame-level metrics, chosen to probe semantically orthogonal axes of similarity.

A. CLIP-based similarity

CLIP similarity captures high-level content agreement. Each frame is encoded with the OpenAI CLIP-ViT-L/14 image tower [11], embeddings are L2-normalized in the joint vision-language projection space, and we report their cosine similarity. Because CLIP is trained on 400M web image/text pairs, this measure is sensitive to scene-level subject and setting but largely indifferent to low-level pixel statistics, making it appropriate for verifying that a generator preserves the meaning of its conditioning signal.

B. Perpetual similarity

LPIPS [12], the Learned Perceptual Image Patch Similarity, occupies an intermediate level. Features are extracted from a fixed AlexNet backbone [13], rescaled by learned linear weights calibrated against human two-alternative-forced-choice judgments on the BAPPS dataset, and the resulting weighted ℓ_2 distance is averaged across spatial locations. LPIPS tracks human similarity judgments substantially more closely than SSIM or PSNR, and is the same family of perceptual constraint

Glaze enforces during cloak optimization [14]. Reporting it therefore measures fidelity against the budget the defense is explicitly engineered to respect.

C. Artistic style representations

Gatys-style loss [15] targets the style axis: feature maps are extracted from VGG-16 [16] at the `relu1_2` / `relu2_2` / `relu3_3` / `relu4_3` layers, the standard perceptual-loss layer set introduced by (author?) [17], channel-wise Gram matrices $G \in \mathbb{R}^{C \times C}$ are formed and normalized by the spatial-channel product $C \cdot H \cdot W$, and the mean-squared difference between corresponding Gram matrices is summed across the four layers. Because Gram matrices encode pairwise channel co-activation statistics independently of spatial position, this metric isolates texture and brushstroke fingerprints from subject content, and remains the canonical similarity criterion in the neural style-transfer literature.

We decided against collapsing the three measures into a single composite, as each probes a distinct axis: content (CLIP), perceptual fidelity (LPIPS), and style (Gram). A successful style cloak should depress style-loss similarity while leaving CLIP similarity largely intact. A perceptually visible cloak would additionally inflate LPIPS. Reporting all three therefore enables a more holistic understanding of generator behavior than any single statistic in isolation. To control for frame-rate and resolution mismatches between the reference and the generation, both clips are uniformly sub-sampled to $N=16$ frames and each frame is resized to 224×224 before evaluation; per-frame metrics are then averaged over the resulting paired frames. We use ImageNet normalization for VGG-based extraction and rescale inputs to $[-1, 1]$ for LPIPS, matching the conventions used during the respective backbones' training. Because Gram-matrix distances span several orders of magnitude in absolute value due to the $1/(C \cdot H \cdot W)$ normalization, we report style-loss results on a $\log_{10}$ scale throughout, which we found to better expose differences across conditions than the raw values.

VI. RESULTS

The results in Table II measure the similarity between videos generated from unprotected reference images and videos generated from the Glazed reference images. Figure 2 shows an example workflow for our VideoGlaze testing process for one of our test animations. Across all three metrics, the two video sets are highly similar: CLIP similarity ranges from 0.8407 (Dynamite) to 0.9705 (Leprechaun) with a mean near 0.926, LPIPS distances are mostly below 0.25 with only Dynamite

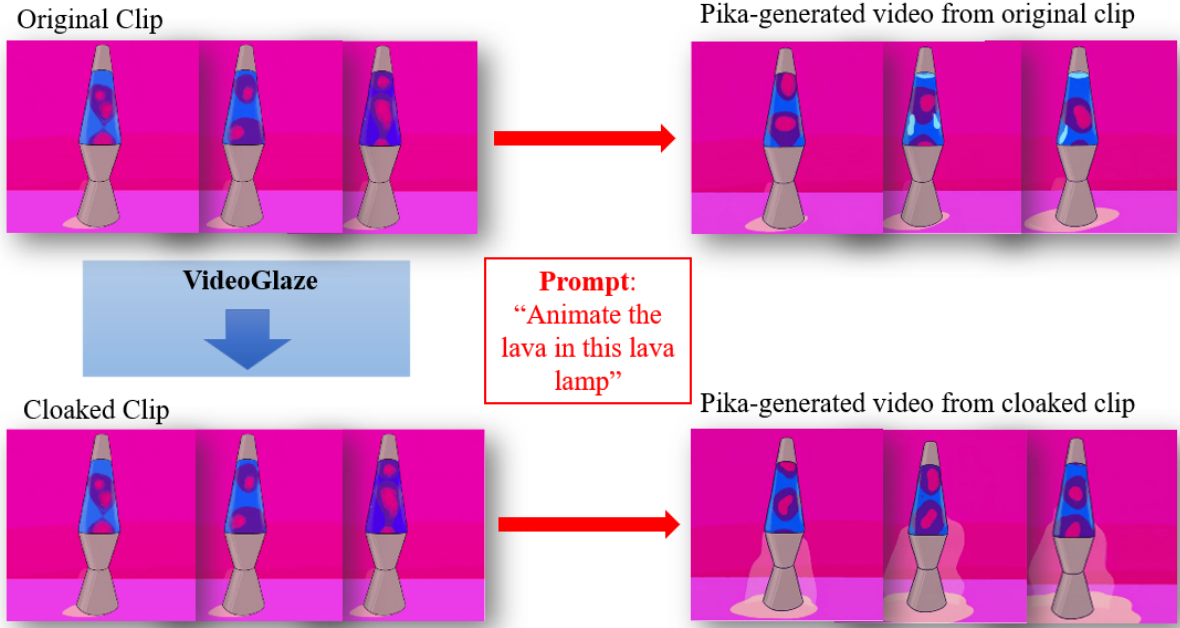


Fig. 2: Example of a VideoGlaze test for the "lava lamp" GIF. **Left:** the original "lava lamp" GIF is cloaked with VideoGlaze, applying Glaze to each individual frame. 3 frames out of 21 in are depicted to show the general video flow. **Right:** Frames from the original "lava lamp" GIF and cloaked GIF are uploaded to Pika with the prompt "Animate the lava in this lava lamp." 3 frames again represent the full GIF.

Test Video	CLIP	LPIPS	Style
Lava lamp	0.9541	0.1110	5.80×10^{-4}
Eye	0.9252	0.1850	1.80×10^{-5}
Dynamite	0.8407	0.4203	2.74×10^{-4}
Leprechaun	0.9705	0.1313	5.00×10^{-6}
Waves	0.9265	0.2294	1.20×10^{-5}
Turtles	0.9366	0.3489	2.05×10^{-4}

TABLE II: Per-clip similarity metrics between generated outputs and reference clips. CLIP similarity (higher value = more similar) measures content agreement in the CLIP-ViT-L/14 embedding space; LPIPS (lower value = more similar) measures perceptual fidelity via learned AlexNet features; style loss (lower value = more similar) is the summed mean-squared Gram-matrix distance over VGG-16 layers `relu1_2`, `relu2_2`, `relu3_3`, `relu4_3`, reported on a $\log_{10}$ scale.

(0.4203) and Turtles (0.3489) showing substantial perceptual divergence, and style loss is uniformly small in absolute terms, ranging from 5×10^{-6} to 5.8×10^{-4} . The high CLIP scores are unsurprising — Glaze is engineered to preserve content while disrupting style,

so the subject and scene should survive cloaking, but the consistently low LPIPS distances and small Gram-matrix discrepancies indicate that the cloak failed to meaningfully perturb either the perceptual fidelity or the texture statistics of the generated video relative to its uncloaked counterpart. The two prompts with elevated LPIPS (Dynamite, Turtles) are also prompts involving large-magnitude motion, and the higher frame-to-frame divergence on these examples may be attributed to normal variation in the video generation method than the cloak itself. Even on the style axis, where a successful style cloak should produce the largest gap, the absolute distances remain significantly smaller than the perturbations the metric is capable of registering on stylistically dissimilar images.

VII. DISCUSSION, LIMITATIONS, AND FUTURE DIRECTIONS

A. Discussion

Our experiment was designed to help independent animators and digital artists in order to protect them from unauthorized style replication. We hope that this tool will help artists that don't have legal resources to pursue copyright claims against large AI companies to be able to keep their brand identity intact. Since text-to-video generation tools are becoming more accessible,

we hoped to be able to address this vulnerability by extending image based tools to video as well. Even though our project showed mixed results and effectiveness, we aimed to show that artists do not need direct access to AI model implementations to be able to take preventative action against style mimicry.

B. Problems Encountered

We originally planned to use text-to-video AI generation tools as well as include the fully cloaked videos as reference. However, most generators that we found online with this capability were behind paywalls so we modified our plan to focus on potential bad faith actors using individual frames from cloaked videos. This is still an important use case for protection of animation videos online. We also set out to include a human evaluation metric as well as empirical tests. However, looking at the generated GIFS and given our results largely did not show that VideoGlaze successfully prevented style mimicry, we excluded this metric. Finally, time limitations prevented us from improving VideoGlaze following our unsuccessful initial results, however we address this possibility in VII-D.

C. Limitations

Given none of the cloaked test videos successfully prevented style mimicry, we can discuss many limitations to our experiment. We focused on shorter, simpler animations because our goal was to create and show proof of concept for VideoGlaze rather than perform a rigorous evaluation. However, the simplicity of these animations may be part of the reason VideoGlaze failed, as more complex artistic styles would be more identifiable by Glaze and able to transform into a new target style. This especially relates to the size of the image, as smaller images naturally have a lower perturbation budget given high perturbations would have a greater effect on distorting the image to the human eye.

Additionally, although we attempted to find newer videos, Pika does not share their training data nor model parameters so it's possible that one or more of our test videos was included in their training data. Although we theorized this would not affect VideoGlaze's effects, given the unsuccessful results we can't rule it out as a possibility.

Few text-to-video AI generators exist that also take in style reference videos rather than just images. Because of this, we were limited to use only one AI testing tool, Pika. However, more tools that do this could emerge in future, especially as companies, VFX studios, and animation studios shift toward integrating AI into their products.

D. Future Directions

There is a lot of potential for VideoGlaze and we hope to expand this tool one day to make it more effective. Most importantly, future experimentation would involve testing with more complex animations, both visually and temporally. Since our current evaluation focused on short clips, we were not able to accurately how well our approach scaled to longer sequences. Furthermore, expanding the dataset to include more photorealistic and detailed image styles would provide a better understanding of how VideoGlaze performs across different artistic domains.

Another promising direction for VideoGlaze is to test it against a wider range of video generation tools. Tools like video extension and various models may represent more realistic threat models. Unfortunately, many of these tools were not included in our experiments due to paywall conflicts and access issues. Additionally, we believe future work can be done to explore stronger and more adaptive Glaze perturbations at the frame level. We hypothesize that this might potentially increase robustness against style extraction.

VIII. CONCLUSION

Through this project, we were able to create a product that extends image based anti-style mimicry defenses to video by applying a software tool called Glaze. Our ultimate goal was to evaluate whether a defensive technique originally designed for static images can be adopted to video in order to protect animated content from text-to-video generation models. Through empirical evaluations, we were able to deduce that while our experiment preserves high level content from the videos, which is reflected by the high CLIP similarity scores, it does not significantly disrupt the style in generated outputs. The low LPIPS distances and minimal differences in Gram-based style metrics demonstrate that current generation models seem to be unaffected by the applied perturbations and are still able to reproduce stylistic characteristics from cloaked inputs.

Overall, these findings highlight a key limitation when it comes to directly transferring image based adversarial defenses to the dynamic. While we tried our best to optimize temporal gating and warm starts, it ended up effectively reducing the strength and diversity of perturbations needed to disrupt style learning amongst frames. Our results suggest that video generation models may rely on some different representations of style compared to image models, making them less susceptible to cloaking techniques. Despite this, our experiment shows the proof of concept in terms of feasibility for a user controlled, proactive defensive tool. We believe that future work should explore this idea further and improve

efficiency for longer videos and evaluate an even wider range of generative models.

IX. SELF-EVALUATION

Srinidhi and Reed initially pursued different ideas for their midterm project and were looking to pivot to a group project that interested them. As Gillian introduced her midterm paper on data privacy, the rest of us became intrigued and even more so when she pitched the idea of applying it to text-to-video models.

A. Srinidhi

I worked on a variety of tasks over the course of this group project. Starting from its conception, I worked with teammates to coordinate logistics on team meetings and made sure to take notes during brainstorm sessions in order to create actionable items for the next meeting. In terms of contributions towards the project, I worked on gathering animation clips for the dataset and researching and implementing the evaluation metrics for determining the success of our project. I also worked on a large portion of crafting this paper.

This entire project was a learning experience for me since I had no prior knowledge about data privacy or the issues surrounding AI generated art. It was very cool to know that a defensive tool like Glaze existed and it was even more exciting to see it be put to use! While we were brainstorming how we wanted to pursue this project, we also came across another tool called Nightshade, which is a proactive tool (opposite to Glaze which is a defensive tool) that disrupts models and makes art unusable for model training. I found it interesting how those researchers advise artists to proceed with protecting their art (they recommend to use a combination of tools, and if not, just to use Glaze with an optional Nightshade tool).

B. Gillian

This project was inspired by my midterm paper. To that end, I came up with the idea and general process for VideoGlaze, although it went through many modifications between the original concept and our final project. I did the background research for our project, led the crafting of our proposals and progress report, helped sourced GIFs for testing, and found metrics for the evaluation phase. I also created the first half of our demo presentation, and wrote and/or edited most of this paper.

I learned a lot about the challenges of transferring an image-based tool to videos, especially the difficulties in maintaining the frames so that transitions between them are preserved. Also, I had not used Glaze myself before, and I learned how slow it was, and by extension

how slow VideoGlaze was too, which really places a time burden on artists who use it. This might discourage artists who don't want to spend the time applying Glaze or VideoGlaze to all their work. Especially if it were used for an extended animation short film or feature, the time for cloaking would likely be prohibitive. I also learned about how difficult it is to define art style and especially to define when an art style is or is not copied.

C. Reed

I focused on building the code and running evaluations for this project. This included implementing the Glaze algorithm and applying it to video, helping determine the evaluation criteria and implementing them, and setting up the GPU environment to run the models.

Working on this project highlighted how difficult protecting against AI mimicry is. I remember Glaze getting a lot of attention in 2023, and it's surprising how poorly it has generalized to modern models and other use cases like video just a few years later. Glaze targeted the state-of-the-art methods of its day, by models are advancing faster than defences against them can keep up, and there are always private models whose unknown architecture makes them especially difficult to defend against. After working on this I think much more attention needs to be paid to protecting cultural work of artists in the coming years.

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